

Credit Cycles and Credit Crises: Some Measurement Issues and Implications^{*}

Josefine Quast^a, Simon van Norden^b, Marc Wildi^{c,*}

^a*International Monetary Fund*

^b*CIRANO, CIREQ, and HEC Montréal*

^c*Institute of Data Science, Winterthur*

Abstract

Basel III allows national regulators to adjust the capital requirements of financial institutions in response to changes in the fragility of the financial system. This paper provides new evidence on the extent to which estimate of the credit cycle can realistically anticipate systemic banking crises. We show that simple theory-based modifications to filtering measures recently mandated by regulators can have an important effect on predictive power. We also highlight how the limited effectiveness of infrequent interventions is consistent with the move to “positive neutral” capital buffers adopted by several nations.

Keywords:

Basel III, Countercyclical Capital Buffer, Banking Crises, Credit Cycles, Bandpass filters, Macroprudential regulation,

^{*}The authors would like to thank ...

^{*} The views expressed in this paper are those of the authors and do not necessarily represent the views of the IMF, its Executive Board, or IMF management.

^{*}Corresponding author

Email addresses: simon.van-norden@hec.ca (Simon van Norden), wlmr@zhaw.ch (Marc Wildi)

URL: www.svannorden.org (Simon van Norden)

1. Introduction

The link between credit expansions and financial instability traces back to Minsky (1977) and Kindleberger (1978), and was later formalized by Bernanke et al. (1999) through the financial accelerator, linking collateral constraints to the amplification of shocks and financial fragility. One of the key accompanying empirical lessons from the Global Financial Crisis (GFC) has been that credit booms often precede systemic banking crises (Reinhart and Rogoff, 2009; Schularick and Taylor, 2012). Since then, a vast literature has explored whether fluctuations in credit can serve as reliable early warning indicators (EWIs) for such crises.

To incorporate these insights into policy frameworks, the implementation of Basel III has been an important response to the GFC. Since it requires operational metrics to guide macroprudential policy decisions, the associated literature evolved around two core questions: how should credit cycles be measured and how should their predictive content for financial crises be evaluated? Both remain crucial since mismeasured credit cycles, on the one hand, could yield miscalibrated countercyclical capital buffers (CCyBs) and overstated accuracy of crisis predictions, on the other hand, may lead decision makers to underestimate uncertainties around modelling choices, giving them a false sense of security in getting the signal and timing right to react to vulnerabilities.

Building on Borio and Lowe (2002), BCBS (2010), Drehmann et al. (2011), and ESRB (2014) formalize the credit-to-GDP gap as a key indicator for excessive economy-wide leverage—defined as the deviation of the credit-to-GDP ratio from its long-run trend as estimated from a one-sided Hodrick–Prescott (HP) filter with a very large smoothing parameter—to serve as a reference for setting the CCyB. Competing indicators are evaluated using Receiver Operating Characteristic (ROC) frameworks, with the area under the ROC curve (AUROC) summarizing the trade-off between missed crises and false alarms, i.e. Type I and Type II errors. Explicit policy criteria are defined as well: appropriate measures must signal sufficiently far in advance, remain elevated throughout the build-up, and avoid excessive false alarms that could undermine policy credibility (BCBS, 2010; Drehmann and

Juselius, 2014).¹

Contribution: In this paper, we frame the construction of the credit-to-GDP gap as a frequency-domain filter design problem to make the implicit choices embedded in existing filters explicit and subject to principled optimization. Using a mean-squared-error-based (MSE) direct filter approach (DFA), building on Wildi and McElroy (2016, 2019), we explicitly target the policy-relevant frequency band associated with medium-term credit cycles and construct optimal one-sided filters given each series’ spectral properties. This makes the trade-offs among *accuracy* (adequate size measurement), *smoothness* (rejecting noise outside of the target bands), and *timeliness* (timely detection of turning points) explicit and quantifiable. For our context, filters that suppress low frequencies like the HP filter tend to lead crisis peaks but generate spurious cycles, while filters that preserve lower frequencies, such as DFA and the Hamilton filter, track credit growth more accurately but lag genuine turning points. We show that this trade-off is governed by tunable filtering features—the zero-frequency amplitude and the passband width—and evaluate its consequences for predictive performance.

Limitations to Measurement and Prediction: Despite its prevalence, Basel’s approach to measuring the credit cycle remains subject to two main critiques: the use of ad hoc, ‘one-size-fits-all’ filter settings and the introduction of spurious cyclical dynamics (see, e.g., Phillips and Jin, 2015; Hamilton, 2018; Schöler, 2018, 2020). We show that the origin of the latter is tied to a fundamental spectral property of the HP filter: when applied to levels, it behaves as a bandpass filter that suppresses the zero-frequency component of the series entirely. This forces the estimated gap to oscillate around zero—exhibiting mechanical mean reversion even when credit growth is persistently elevated.² Hartwig et al. (2021) document one consequence: HP gaps signal systemic risk incoherently across countries, with high gap values indicating risk in some economies and low values in others, impeding a consistent

¹Brave and Lopez (2019) showcase how to use crisis probabilities to inform macroprudential policy and directly link early warning properties to the calibration of CCyBs.

²Under white noise, this property generates a sign change in the gap roughly every 4.5 years, producing a spurious cycle approximately every 9 years regardless of underlying credit dynamics.

economic interpretation.³

Moreover, standard evaluation practices likely overstate the predictive performance available to policymakers in real time. First, the Basel gap was selected in the aftermath of the GFC, relying heavily on the preceding credit boom and validating its predictive power requires genuinely new crisis data not used in its design (van Norden and Wildi, 2015; Quast, 2026). Second, AUROC-based evaluations using expanding-window filters average across very different filtering regimes, with short distorted samples for early observations and longer more accurate ones for later observations, making assessments increasingly misleading as longer-sample regimes dominate over time.⁴ Third, end-point biases and reliance on revised rather than real-time data further overstates performance available to policymakers in real time (Edge and Meisenzahl, 2011; van Norden and Wildi, 2015; Alessandri et al., 2015). Fourth, since credit-to-GDP gaps vary in scale across countries, we show that pooling without adjustment disproportionately weights nations with the largest or most variable gaps, and that rescaling and standardizing gaps can change pooled rankings along the ROC curve and with them the measured AUCs and partial AUCs. These limitations motivate an approach that makes implicit filter choices explicit and assesses their consequences for predictive performance transparently.

We find that DFA achieves the highest overall predictive accuracy, outperforming the HP and Hamilton filter and the five-year difference. But along the ROC curve, rankings depend on policymakers' tolerance for false alarms, and there are bounds to what credit gap measures can predict, irrespective of the filtering method, since not all systemic banking crises follow

³They attribute this at least partially to spurious medium-term cycles as discussed in Schüler (2018, 2020). Further, Quast (2026) shows that HP-based approaches utilizing very smooth trend parameters remain susceptible to spurious regression issues and Hartl (2025) show that trend estimates become highly sensitive to the misspecification of persistence when the underlying persistence of the process remains unknown. Other well-documented limitations include procyclicality issues (BCBS, 2010; Repullo and Salas, 2011; Jokipii et al., 2021) or implausible country patterns (ECB, 2017; Baba et al., 2020).

⁴Beutel et al. (2019) find that machine learning methods, despite greater flexibility, do not tend to outperform conventional logit-based models in out-of-sample crisis prediction, suggesting that standard approaches extract the available predictive information fairly efficiently.

credit booms, and not all credit booms end in crises. BCBS (2010) envisioned activating the CCyB rarely. So, for false-alarm rates below, for example, ten percent, the HP filter attains the highest partial AUC. Yet, none of the filters we investigate can deliver an infrequent buffer activation while also providing reliable crisis detection since Hit rates remain below 50% for all four approaches considered in the low false alarm region. This binding trade-off provides a rationale for positive neutral CCyBs which build resilience unconditionally rather than depending on a signal that cannot be made simultaneously timely and selective.⁵

Recognition of measurement and crisis prediction limitations has motivated a shift toward positive such neutral CCyB frameworks, which maintain a standing buffer that can be released in periods of distress rather than relying exclusively on predictive build-up signals. By 2025, 17 EEA countries had adopted a positive neutral CCyB, aiming to enhance resilience by building adequate buffers in a timely manner while spreading the costs of accumulation over longer horizons (ECB/ESRB, 2025).

Cross-country Considerations: Customization and nuanced perspectives matter in the credit cycle context, not least since the length of credit cycles tends to vary across economies and over time (Galán, 2019) and macroprudential policy decisions are taken by national policymakers. Although the Basel III guidance envisages a ‘one-size-fits-all’ approach, in practice and especially among transition and developing economies, gap extraction is achieved with a noticeable degree of discretion to navigate structural breaks and financial deepening that pose challenges to the recommended filtering techniques (Wezel, 2019). Beyond idiosyncratic implementation, there is also evidence that the predictive content of Basel III credit-to-GDP gaps is heterogenous across countries (Detken et al., 2014). Beltran et al. (2021) find credit-gap-based early warning indicators to perform better for advanced economies than for emerging market economies and attribute this to fundamentally different crisis-generating mechanisms. Conversely, Chen and Svirydzenka (2021) find that credit gaps remain informative in emerging market economies due to their bank-intermediated structures. This tension in findings can likely be traced back to the different frequency bands both studies

⁵Regulators with higher tolerance for frequent buffer deployment, willing to accept more non-crisis activations in exchange for earlier warnings, would prefer DFA-MSE.

target: while Beltran et al. (2021) evaluate filters following the Basel III tradition, Chen and Svirydzenka (2021) extract cycles at much shorter, medium-term frequencies to capture the more volatile component of credit dynamics in emerging market economies. Both approaches showcase that treating the targeted frequency band as an explicit design choice matters and depends on the question of interest.

Multivariate Considerations: Beyond evaluating credit-to-GDP gaps and their usefulness for predicting systemic banking crises on a stand-alone basis, multivariate approaches have been put forward as well. Behn et al. (2013) and Detken et al. (2014) find that multivariate settings improve the power of predictive models.⁶⁷ Lang and Welz (2017) show that how the gap is constructed also matters for the predictive ability in semi-structural, parametric settings when they construct the credit gap as deviations from a fundamentals-based trend in an unobserved components model. Using a Bayesian VAR to construct credit forecasts based on fundamentals, Afanasyeva (2020) shows that deviations of credit from model-predicted values provide stronger crisis signals than statistical credit-to-GDP gaps. While this strand of the literature demonstrates that systemic risk tends to emerge from interacting imbalances as excessive credit growth tends to coincide with asset price booms or external funding surges, credit-to-GDP continues to be identified as the ‘best single indicator’ for predicting systemic banking crises. In this paper, we maintain the focus on univariate credit-to-GDP gaps and their performance in crisis prediction for the following reasons: first, institutionally, the Basel III guidance places considerable emphasis on univariate, easy-to-implement measures that are easy to compare across countries. Second, improving credit gap measurement benefits practitioners regardless of whether the gap is used on a stand alone basis or as an input into broader (semi-) structural settings.

Outline: The remainder of the paper proceeds as follows. Sections 2 and 3 lay out

⁶While Behn et al. (2013) find asset prices and global factors to be particularly useful, Detken et al. (2014) find that including debt service-to-income, current account-to-GDP, and real equity price growth improves predictions in a multivariate setting.

⁷To evaluate, Detken et al. (2014) combine AUROCs/partial AUROCs with loss functions reflecting the preferences of policy makers.

details on the data used in this study as well as on commonly-used univariate measurement concepts for credit-to-GDP gaps. The latter also includes descriptions of the DFA. Section 4 evaluates the predictive performance of the various credit cycle measures, before section 5 concludes.

2. Data

Our quarterly data set relies on BIS credit statistics from the long series on credit to private non-financial sectors database as used in the BIS-related literature. We consider credit-to-GDP ratios that measure credit extended to the private non-financial sector, scaled by GDP. These series cover 44 middle and high-income countries from as early as 1961Q1 up to 2025Q1, but only 14 countries have data reaching back to 1970Q1 and only 22 reach back to 1980Q1.

The other element that we require for our empirical analysis is a set of dates for the start of systemic banking crises. The most commonly used dates are those provided in Laeven and Valencia (2018, 2020), who identify systemic banking crises from 1970 to 2017. We consider Nguyen et al. (2022) as well, who provide information on crises dates until 2019. For the countries considered in our study, however, no systemic banking crises is recorded in the additional years. To bridge the gap to available credit data and in absence of events since the end of Laeven and Valencia’s (2020) database, we extend the data set with zeros until 2025Q1. A few countries suffered more than one banking crisis, as the table below shows, while several saw no systemic crises. The relative frequency of crises also seemed to vary, with crises appearing more frequently in the latter part of the sample. As we will see below, one implication of this is that the overall predictive performance of the various measures we examine may be sensitive to the sample period examined.

3. Measuring the Credit Cycle

In this section, we review and compare different approaches to estimating credit gaps by comparing their properties as linear filters. We compare their filter weights, cyclical

properties as well as the gap estimates they produce for some major economies. Throughout, we distinguish the behaviour of “ideal” 2-sided filters (which may require infinite numbers of future and past observations) and “feasible” 1-sided filters (restricted to a finite number of current and lagged observations, such as may be available to decision makers.) We apply the Direct Filter Approach (DFA) of (Wildi and McElroy, 2019) which constructs optimal feasible filters for a general class of ideal linear filters. We also relate their properties as applied to the level of a series to those implied for the first-difference of a series.

We briefly review the differing properties of the two-sided and one-sided HP gap filters, of the Hamilton as well as of the ordinary five-year difference filters, before turning to consider a simple feasible DFA filter. We show that the latter provides a sharp cut-off at the target frequency and that it behaves as a band-pass filter on the level of the series although specified as a low-pass (trend) filter in first-differences. We argue that the credit gaps estimated with the HP filter are contaminated by a tendency to produce spurious cycles (such as we find, e.g., for Great Britain and Japan) while in theory this may be mitigated by the DFA filter.

3.1. Classic Designs

3.1.1. The HP Filter

The Hodrick-Prescott (HP) filter can be interpreted as an optimal MSE-signal extraction filter for the trend in the smooth trend model, see Harvey (1989). Conceptually, this results in ‘implied’ models for the cycle and the trend, such that applying the HP filter results in MSE optimal estimates. In this framework, the bi-infinite symmetric (i.e. “two-sided”) expansion of the filter is obtained as

$$(\gamma_{|k|} B^k)_{|k|<\infty} = \frac{1}{1 + \lambda(1 - B)^2(1 - B^{-1})^2} \quad (1)$$

where $\gamma_{|k|}$ is the filter weight at a lead or lag of k , λ is a ‘smoothing’ hyperparameter and B, B^{-1} are backward and forward operators.⁸ In finite samples, the filter behaves differently toward the boundaries of the data, where the symmetry is lost and an exact finite sample

⁸The implicit data-generating process follows an ARIMA(0,2,2)-specification whose MA-coefficients are determined by λ , see (McElroy, 2008).

representation of the concurrent or “one-sided” HP trend filter, denoted as $b_k^{HP-trend}$, can be found in (McElroy, 2008) and Cornea-Madeira (2017).

Cycle analysis concerns the deviation of an indicator from its trend which is sometimes referred to as the gap. Accordingly, the two-sided HP^{gap} filter is defined by

$$\gamma_k^{gap} \equiv \begin{cases} 1 - \gamma_0^{HP-trend} & k = 0 \\ -\gamma_k^{HP-trend} & , \quad k \neq 0 \end{cases} \quad (2)$$

and b_k^{gap} is defined analogously in terms of $b_k^{HP-trend}$. Fig.(1) displays filter coefficients (top) and credit-cycles (bottom), for $\lambda = 400000$. The gap filters place most weight on the current observation. While the one-sided weights on past observations are individually small, they decay slowly and satisfy $\sum_{k \geq 0} b_k^{gap} = 0$, implying that changing the level of the series by a constant has no effect on the calculated gap.⁹

3.1.2. The Hamilton Filter and Difference Filters

Hamilton (2018) defines the cycle as the residual from a direct h -step-ahead $AR(p)$ forecast,

$$y_t^{Ham} = x_t - (c + \sum_{k=1}^p a_k x_{t-(h-1)-k}). \quad (3)$$

In the case of credit/GDP ratios, he suggests using $p = 4$ and $h = 20$.¹⁰ Subsequent work by (Hamilton and Leff, 2020) and (Drehmann and Yetman, 2021) note that instability of the implied filter coefficients in countries with relatively short data series (e.g. 20-30 years) may be alleviated by either using pooled estimation methods to impose common $\{a_k\}$ across all countries, or by simply using a 5-year difference:

$$y_t^{Diff} = x_t - x_{t-20},$$

⁹One can show that the HP-gap removes linear functions.

¹⁰The 5-year forecast horizon that he suggests for credit/GDP ratios differs from the 2-year horizon he suggests for business cycles and is intended to capture the longer durations of credit cycles.

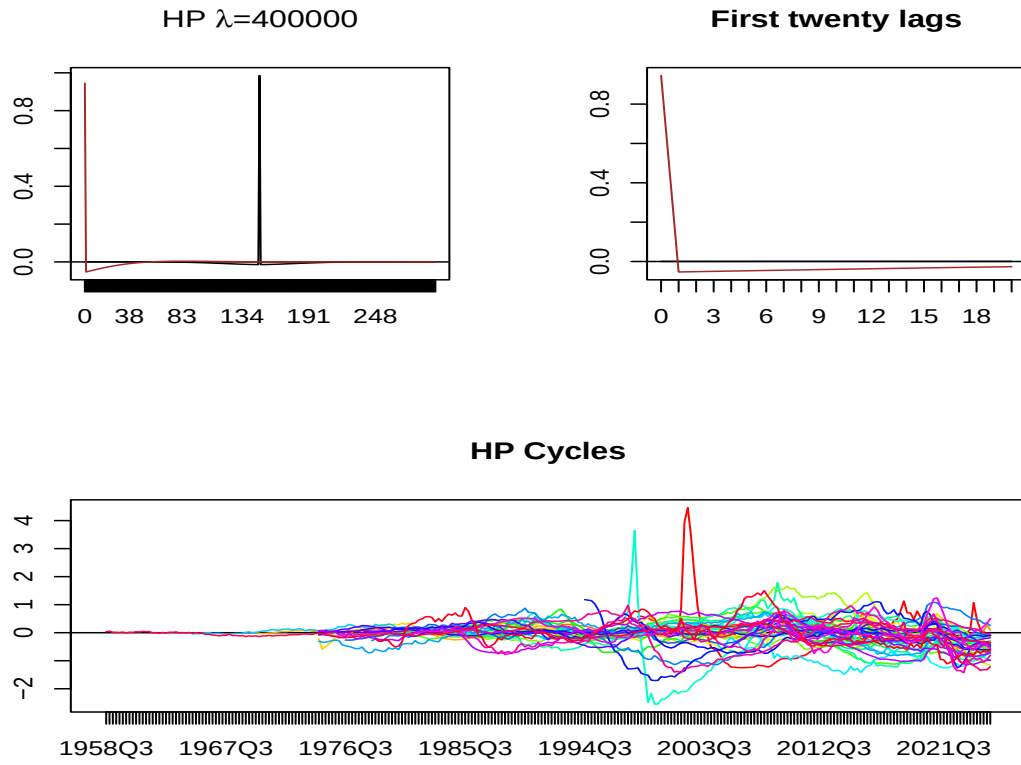


Figure 1: HP-Gap Filter Weights and Estimates, $\lambda = 400,000$.

Top left: Two-sided (black) and one-sided (brown) filter weights.

For clarity, weights for the two-sided HP (black line) are shifted to the right.

Top right: Weights for the one-sided HP-Gap filter

Bottom: Estimated gaps (cycles). Cycles are scaled to unit variance for comparison.

which is the special case with $c = a_2 = a_3 = a_4 = 0$ and $a_1 = 1$. We examine both below. For the Hamilton filter, we used panel estimation with fixed effects c_j and an expanding-window fit.¹¹

3.2. Direct Filter Approach (DFA)

DFA is a general forecasting framework formulated in the frequency domain (see McElroy and Wildi, 2019). Let the target

$$z_t = \sum_{k=-\infty}^{\infty} \gamma_k \Delta x_{t-k},$$

i.e., the result of applying an acausal, square-summable filter $\{\gamma_k\}$ (such as the HP) to Δx_t .¹² ¹³ While z_t may be a desirable target, it is infeasible in practice since it depends on (unknown) future values $x_{t-k|k<0}$. Instead, we seek a causal (i.e. one-sided) finite-length predictor applied to Δx_t

$$y_t = \sum_{k=0}^{L-1} b_k \Delta x_{t-k}$$

that tracks z_t in an optimal sense. As a baseline, consider the MSE norm

$$MSE = \frac{1}{2\pi} \int_{-\pi}^{\pi} |\Gamma_{\gamma}(\omega) - \Gamma_b(\omega)|^2 dH(\omega) \quad (4)$$

where $\Gamma_{\gamma}(\omega) = \sum_{k=-\infty}^{\infty} \gamma_k \exp(-ik\omega)$ and $\Gamma_b(\omega) = \sum_{k=0}^{L-1} b_k \exp(-ik\omega)$ are the transfer functions of γ and $\mathbf{b}$ and $H(\omega)$ is the spectral distribution.¹⁴¹⁵ We may further decompose the

¹¹We also investigated imposing a unit-root restriction ($\sum_{k=1}^p a_k = 1$ on (3)). While this substantially improved the performance in short samples, pooled estimation and the 5-year difference performed better still. Similarly, while we found some statistical evidence that $p = 20$ fit the dynamics better in full samples, the estimated gaps were largely similar to those with $p = 4$ but less stable.

¹²Below in our applications to credit cycles, we find that credit ratios are typically non-stationary but that their differences are stationary. Here, for ease of exposition we assume that the series being analyzed is stationary and denote it Δx_t . The mapping back to non-stationary levels is discussed below.

¹³The value of an *acausal* filter at t may depend on current, past and future values of x_t . In contrast, the value of a *causal* filter at t may only depend on current and past values of x_t .

¹⁴We use “target filter” interchangeably for the time-domain coefficient γ and their frequency-domain representation $\Gamma_{\gamma}(\omega)$, and likewise for $\Gamma_b(\omega)$.

¹⁵If the process is stationary and the Wold coefficients decay sufficiently fast, then $dH(\omega) = h(\omega)d\omega$ where $h(\omega)$ is a continuous spectral density. Extensions to nonstationary processes are given in (Wildi and

transfer functions

$$\Gamma_\gamma(\omega) = |\Gamma_\gamma(\omega)| \cdot e^{i \arg(\Gamma_\gamma(\omega))} = A_\gamma(\omega) \cdot \Phi_\gamma(\omega) \quad (5)$$

and similarly for $\Gamma_b(\omega)$ into amplitude $A_\gamma(\omega) \equiv |\Gamma_\gamma(\omega)|$ and phase $\Phi_\gamma(\omega) \equiv \arg(\Gamma_\gamma(\omega))$.

To distinguish trends from cycles, we consider a one-sided filter constructed to approximate the ideal low-pass target (applied to Δx_t)

$$\Gamma^{id}(\omega) = I_{\{|\omega| < \nu\}},$$

where I_Ω denotes the indicator function. We choose the cutoff ν to emphasize fluctuations with periods of at least 30 years—i.e., below the frequencies associated with credit cycles (for example, see Drehmann et al.(2010)) and well beyond those typically targeted by HP or Hamilton filters (for example, see Hamilton (2018)). Substituting $\Gamma^{id}(\omega)$ for $\Gamma_\gamma(\omega)$ into (4) and minimizing with respect to $\{b_k\}$, DFA delivers the corresponding optimal one-sided estimate of the ideal two-sided trend. For the spectral density $h(\omega) = dH(\omega)/d\omega$, one can either fit a parametric model to the data (like that used in (Hamilton, 2018)) or, alternatively, exploit the fact that (after differencing) credit/GDP ratios are close to white noise. In the latter case, it is natural to assume a flat spectrum $h_{\Delta x}(\omega) = C > 0$.¹⁶ The resulting filter requires no estimation, which can be particularly advantageous when the available sample length is too short to provide reliable information about very long-run dynamics (see, for example, (Hamilton, 2018), especially Section IV.B).

Fig. (2) shows the filter coefficients $b_k^{DFA-diff}$ based on the MSE criterion (4), under a white-noise spectrum for Δx_t . The cutoff ν governs the length of the damped cyclical pattern in the coefficients (top panel). The estimated low-frequency components in the bottom panel appear to be stationary because we applied the trend filter to differences Δx_t .

McElroy, 2016).

¹⁶To account for potential seasonality, particularly in GDP, one can consider yearly differences $x_t - x_{t-4}$. This would modify the spectral density to $h(\omega) = \sum_{k=0}^3 \exp(-ik\omega)$, reflecting the equally-weighted MA(4) structure inherent in the yearly difference filter.

As shown below, DFA cycles are typically smoother than HP cycles, with fewer spurious pre- or post-crisis rebounds, reflecting the two filters' different treatment of low-frequency components.

3.3. Filter Characteristics and Cycles: First Differences vs. Data in Levels

To compare the original HP-Gap, defined on levels x_t , and DFA, defined on first differences Δx_t , we can map the DFA low-pass filter back to the level series using the transformation

$$b_k^{DFA-level} = \begin{cases} b_0^{DFA-diff}, & k = 0 \\ b_k^{DFA-diff} - b_{k-1}^{DFA-diff}, & 0 < k < L \\ -b_{L-1}^{DFA-diff}, & k = L \end{cases} \quad (6)$$

When applied to levels, this transformed filter reproduces exactly the output obtained by applying $\{b_k^{DFA-diff}\}$ to differences:

$$y_t^{DFA} = \sum_{k=0}^{L-1} b_k^{DFA-diff} \Delta x_{t-k} = \sum_{k=0}^L b_k^{DFA-level} x_{t-k}$$

Alternatively, to gain insight, write the one-sided HP-gap filter on levels, denoted $\{b_j^{HP-level}\}$, in first differences as

$$b_k^{HP-diff} = \sum_{j=0}^k b_j^{HP-level}, \quad (7)$$

so that¹⁷

$$y_t^{HP} = \sum_{k \geq 0} b_k^{HP-level} x_{t-k} = \sum_{k \geq 0} b_k^{HP-diff} \Delta x_{t-k}.$$

A similar transformation applies to the Hamilton filter and the 5-year difference. We can therefore compare the HP-Gap, Hamilton and DFA directly in either levels or first differences: although the filter characteristics change under this transformation, the resulting cycles do not.

Fig.(3) compares filter characteristics of HP-Gap, Hamilton, 5-year differences and DFA, when applied in levels and first differences: coefficients of level filters applied to x_t (top

¹⁷Level- and diff-filters remove unit-roots of order two and one, respectively.

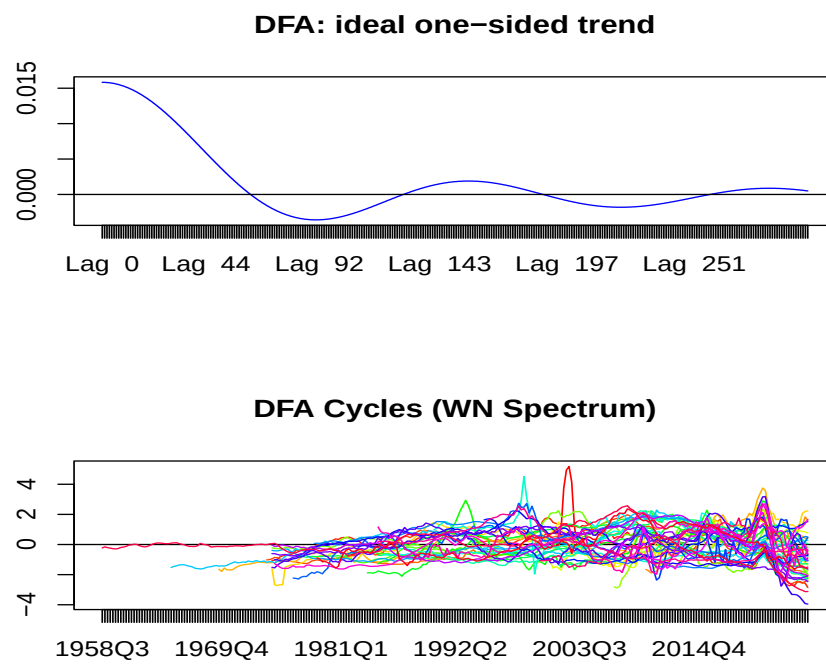


Figure 2: DFA ideal trend with 30 year cutoff and white noise spectrum.

Top: filter coefficients; bottom: DFA cycles obtained from differenced data, scaled to unit-variance.

left) with amplitude functions (top right); coefficients and amplitudes of diff-filters, applied to Δx_t (middle and bottom). When applied to levels x_t , the amplitude functions show that $b_k^{HP-level}$ (brown) is a high-pass filter, $b_k^{DFA-level}$ (cyan) is a band-pass with a single peak around a 30-year period (set by the cutoff ν), and both Hamilton (red) and 5-year differences (orange) display notches at or near multiples of $\pi/10$, i.e., $2\pi/h$, eliminating seasonal frequencies at $\pi/2$ and π . In first differences Δx_t , $b_k^{HP-diff}$ (grey) becomes band-pass while $b_k^{DFA-diff}$ (blue), Hamilton (violet) and 5-year differences (green) are low-pass (non-vanishing amplitudes at $\omega = 0$).¹⁸ The figure reveals considerable scaling disparities between DFA-diff and all other filters, the latter tending to substantially overweight low-frequency components (DFA-diff is therefore shown separately in the bottom panels). Of particular note, DFA-diff displays a sharp attenuation at the 30-year cutoff, whereas HP-diff, Ham-diff, and Diff-diff exhibit more gradual roll-offs toward higher frequencies.

Fig.(4) compares cycles for selected countries with sufficiently long histories that also experienced crises.¹⁹ This highlights a fundamental tradeoff: while HP tends to lead DFA ahead of crises, its bandpass characteristic — namely, the vanishing amplitude at frequency zero — introduces a tendency to emphasize additional, potentially spurious peaks, either before the actual crisis (IT, KR, and GB) or after it (JP, KR), which may distort crisis detection. Hamilton sits between the two extremes.

To conclude our analysis, Figure (5) compares DFA and HP for the US. The DFA is directly interpretable in terms of the quarterly growth of Credit-over-GDP, as the amplitude of DFA-diff sits close to one throughout the passband. HP, by contrast, does not lend itself to such a straightforward interpretation, since the amplitude of HP-diff reaches a maximum of 17.25 at its peak, see Figure (3) (middle right). This difference in scales is clearly visible in the bottom-left panel of Figure (5) and necessitates rescaling before meaningful comparisons

¹⁸Cycles are unchanged when moving between levels x_t and differences Δx_t , because the transformations (6) and (7) offset the change.

¹⁹Comparison plots for each of the countries in our data set may be found in the Appendix.

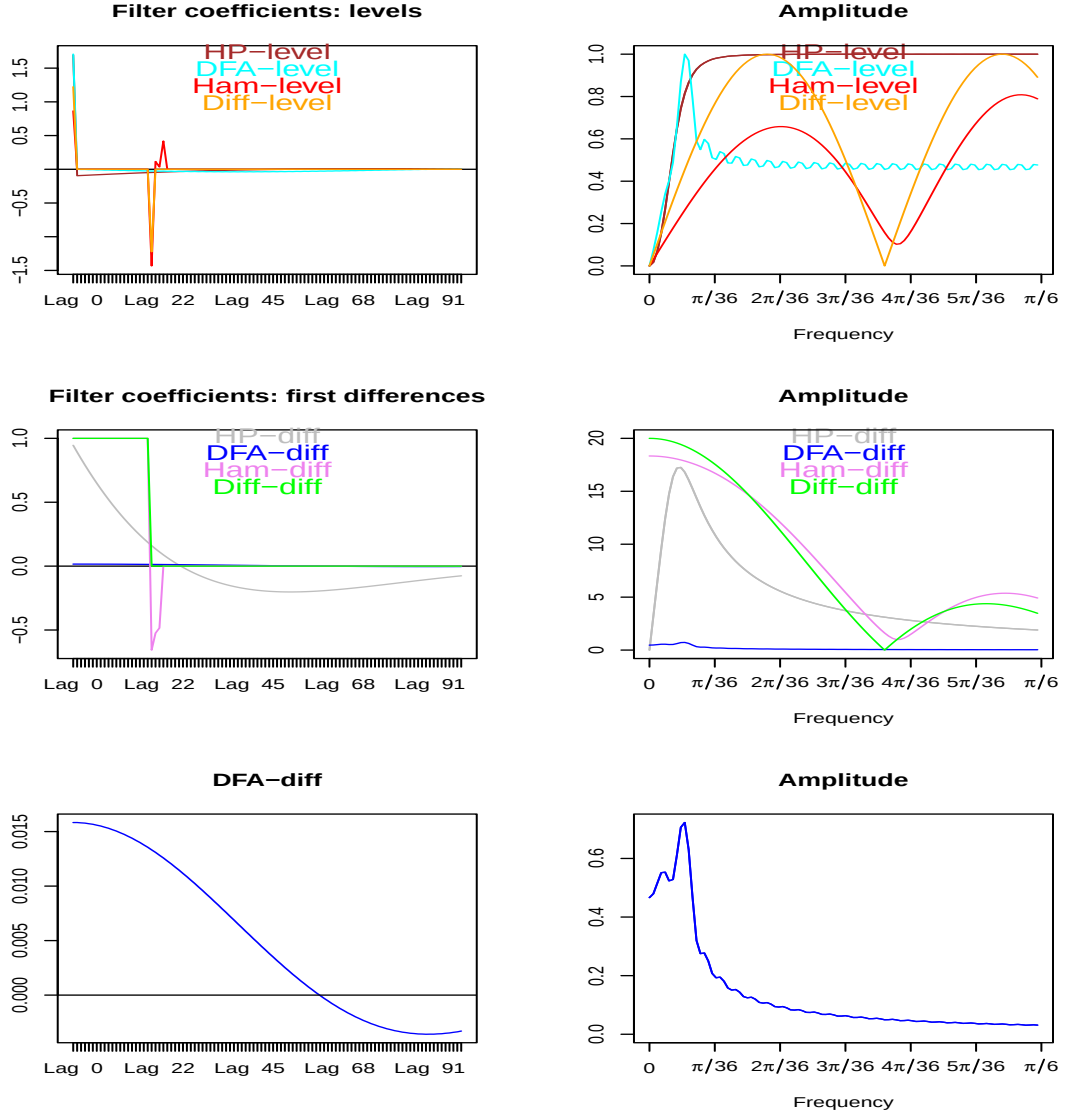


Figure 3: HP, Hamilton, 5-year differences, and DFA in levels and first differences: filters applied to x_t (top) and to Δx_t (middle and bottom); coefficients (left) and amplitude functions (right). Amplitudes are displayed over a narrower passband of width $\pi/6$ in order to highlight the most relevant filter characteristics. To facilitate comparisons in the top panel, filter coefficients are scaled to unit length and amplitudes are normalized to a maximum of one. Diff-filters are left untransformed: owing to scaling differences, DFA-diff is shown in separate panels (bottom). Applying level-filters (top) to x_t yields identical cycles to those obtained by applying diff-filters (middle) to Δx_t .

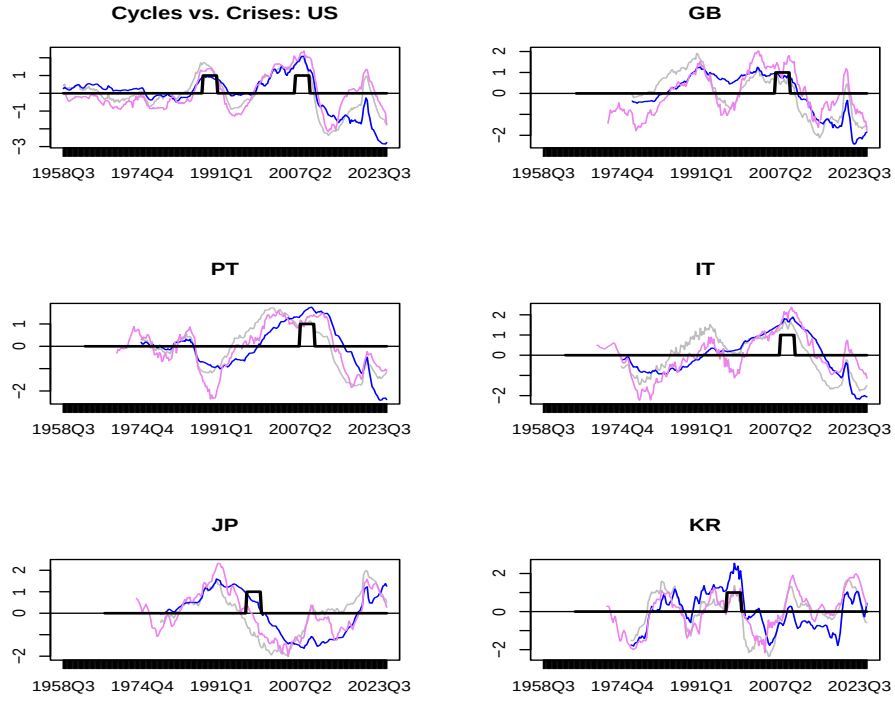


Figure 4: Cycles (scaled to unit variance) for selected crisis-hit countries with long time series: HP band-pass (gray) versus Hamilton (violet) and DFA low-pass (blue). The HP band-pass has a more pronounced tendency to produce spurious pre-/post-crisis cycles. Hamilton and DFA low-pass filters follow growth in x_t ; Hamilton's wider passband lets shorter cyclical movements "leak through", albeit damped.

can be drawn (bottom-right panel). The pronounced upward trend in the series yields positive DFA readings up until the early 2000s and the subsequent second financial crisis.

As shown in the top-right panel, DFA surpasses the first crisis growth-peak (attained in 1987-Q3) as early as 2001-Q2 towards the second crisis (intersection of red and black lines). The HP cycle does not exceed its previous crisis peak until 2006-Q3, five years later than DFA. We argue that deviations from trend, as captured by HP, are inherently more difficult to interpret than growth rates. Furthermore, even though HP tends to lead DFA at true crisis peaks, this behavior does not necessarily translate into earlier breaches of previously critical threshold levels — assuming that such historical levels are considered relevant benchmarks for raising capital buffers.

3.4. Controlling Mean Reversion and the Lead-Cyclical Trade-Off

Mean reversion in a zero-mean process implies that the frequency of sign changes is determined by its dependence structure. In particular, the expected time between consecutive sign changes, is

$$ht = \frac{2\pi}{\arccos(\rho_1)}$$

where ρ_1 is the first-order autocorrelation; see Wildi (2026a), Wildi (2026b).²⁰ Under white noise, ρ_1 is pinned down by the filter, implying²¹

$$ht_{DFA} = 34.8 \text{ and } ht_{HP-Gap} = 18.8.$$

Under white noise, the results suggest that HP-Gap would induce a sign change about every 4.5 years (i.e., a spurious cycle roughly every 9 years), even absent crises. DFA yields about half as many spurious sign changes and cycles. As the mean level in Δx_t , i.e., the drift in x_t , increases, DFA generates progressively fewer sign changes, whereas HP-level (or HP-diff) remains insensitive, maintaining the 4.5-year frequency.

²⁰Explain why Gaussianity holds, i.e., via a CLT argument, even under very strong smoothing filters.

²¹These durations would be longer under positive autocorrelation, but the relative comparison would remain broadly unchanged.

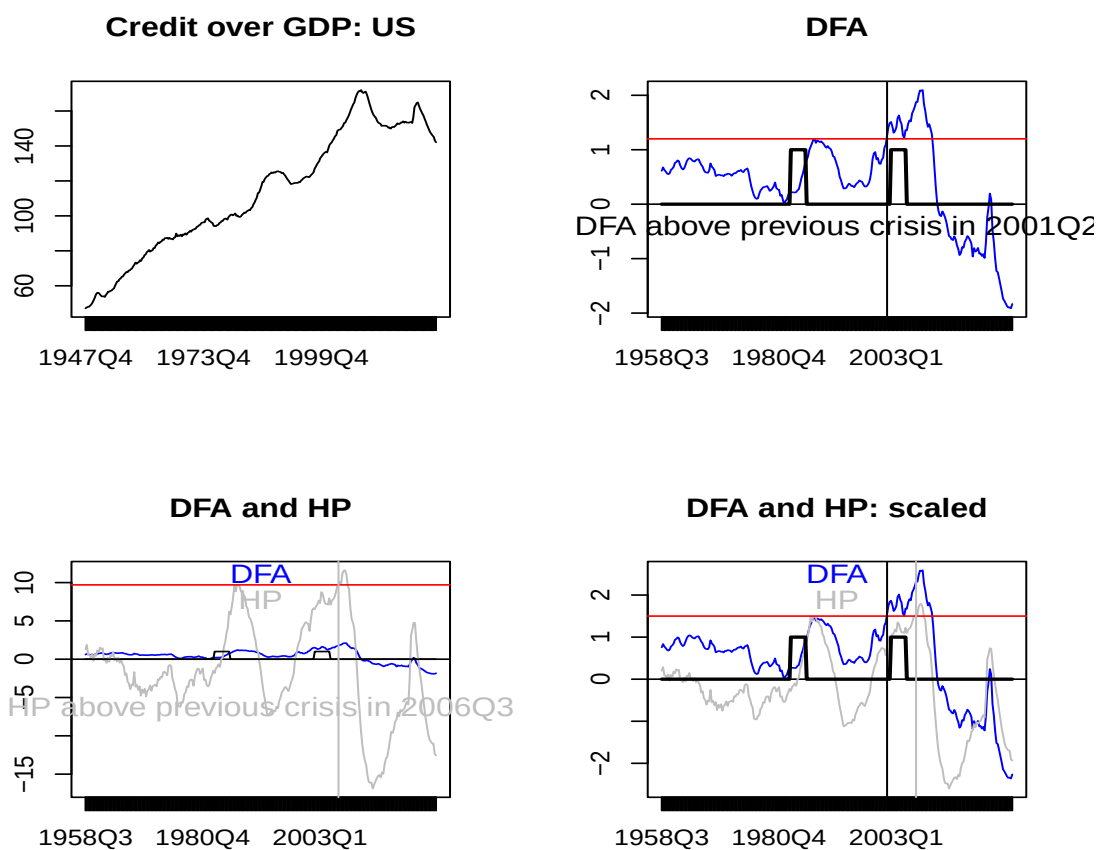


Figure 5: DFA lowpass versus HP bandpass for the US. The top-left panel shows credit over GDP; the top-right panel displays the DFA (blue) and crises (black); the bottom-left panel overlays DFA and HP; and the bottom-right panel presents scaled versions of both. The DFA (top) captures quarterly growth, with the red horizontal line marking the peak growth recorded at the first crisis. DFA surpasses that previous peak level in 2001-Q2 (indicated by the black vertical line), whereas HP exceeds its prior crisis level in 2006-Q3, indicated by the grey vertical line (bottom).

This higher frequency of spurious cycles (stronger mean-reversion) can be linked to the bandpass behaviour of the filter, which damps or eliminates the lowest frequencies. For illustration, consider a backcast of the one-sided HP trend filter $b_{k,Lag}^{trend}$, i.e., the causal length- L estimator

$$y_{t,Lag}^{trend} = \sum_{k=0}^{L-1} b_{k,Lag}^{trend} x_{t-k},$$

which targets z_{t-Lag} (cf. (1)). Setting $Lag = 0$ yields the standard (length L) one-sided HP trend nowcast. To obtain a cyclical component, define the corresponding one-sided gap filter as

$$b_{k,Lag}^{gap} = \begin{cases} 1 - b_{0,Lag}^{trend}, & k = 0 \\ -b_{k,Lag}^{trend}, & 0 < k \leq L - 1 \end{cases}. \quad (8)$$

For $Lag = 0$, $b_{k,Lag}^{gap} = b_k^{HP-level}$, i.e., the gap nowcast. To highlight its key properties, we transform the filter to first differences Δx_t as in (7):

$$b_{k,Lag}^{gap-diff} = \sum_{j=0}^k b_{j,Lag}^{gap}, \quad (9)$$

Fig.6 shows the filter coefficients, amplitude functions, and GB cycles for $b_{k,Lag}^{gap-diff}$ in (9) as a function of Lag , assuming a white noise spectrum. As Lag increases, the filter becomes less cyclical (higher amplitude $A(0)$ at $\omega = 0$, i.e., weaker mean reversion), the shift rises, and the cycle shifts rightward. For sufficiently large Lag , the dominant peak (the maximum in the series) moves from the left spurious cycle to the rightward crisis cycle, aligning with the DFA maximum and the GB crisis. This highlights a trade-off between leading at the true peak (small time-shift: upper right panel in Fig.6) and inducing potentially spurious cyclical-ity, tied to a vanishing (or small) low-frequency amplitude $A(0) = 0$ at $\omega = 0$ (top left panel).

The undesirable first spurious peak in GB (or the additional peaks in the other series) can be further damped by narrowing the passband, as in DFA, where the cutoff ν (30 years) controls passband width. Raising λ in the classic HP-Gap does not resolve this, since HP-Gap's band-pass structure removes the zero-frequency component regardless of λ

(its amplitude vanishes at $\omega = 0$). In summary, the effect of undesirable (spurious) side peaks can be mitigated by (a) allowing zero frequency to pass ($A(0) > 0$: low-pass behavior) and (b) narrowing the passband—at the cost of reduced lead (or greater lag) relative to the ‘true’ crisis peak. The effect of this trade-off on crisis-detection is evaluated in Section 4.

Note that $b_{k,Lag}^{gap}$ as defined in (8) (or its transformed variant (9)) is not the backcast of the original gap filter: the original places most weight on x_{t-Lag} , whereas $b_{k,Lag}^{gap}$ places most weight on x_t . The purpose of (8) or (9) is to formalize the trade-off between cyclicity (measured by the amplitude $A(0)$ at $\omega = 0$; top left panel of Fig.6) and lead (measured by the time-shift: top right panel) in crisis detection: larger $A(0)$ (weaker mean reversion) implies a larger shift (more retardation), and vice versa. A more refined treatment is available within DFA by tuning $A(0)$ at $\omega = 0$ while controlling for the shift; see (Wildi and McElroy, 2019).

4. Crisis Prediction

4.1. Methodology

This section sets out the methodology for evaluating the performance of the various credit cycle measures. We use the ROC (Receiver Operating Characteristic) framework that has been widely used in the previous literature.²² Throughout, we take the perspective of banking regulators operating a CyCB, who try to anticipate the *start* of a systemic crisis. We also assume that they face a binary decision: to set the CyCB at a low (e.g. 0%)

²²Binary variables, such as the crisis indicator used here, may also be modelled using probit or logit analysis to produce probability forecasts. By minimising the mean squared error (MSE) of the forecast, however, those methods weigh crisis and non-crisis outcomes equally. Given the relative rarity of crises, this places much more weight on avoiding false alarms (spurious crisis warnings) than on avoiding missed alarms (failure to warn of a crisis.) The ROC framework instead provides regulators with all possible trade-offs between both types of errors.

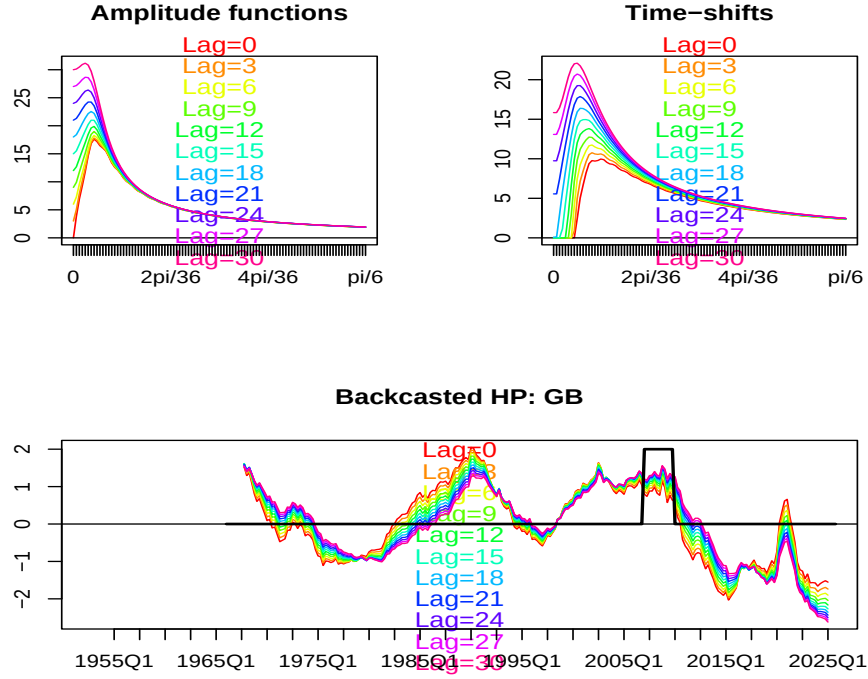


Figure 6: Backcasts: unit-scaled HP-cycles (bottom) as a function of Lag , with the GB crisis in black. Amplitudes and shifts are shown over a narrower passband of width $\pi/6$ to highlight the relevant filter features. As Lag increases, the amplitude at $\omega = 0$ rises (weaker mean reversion), the shift increases, and the cycle moves rightward. The cycle's maximum moves from the left, spurious peak at $Lag = 0$ to the right crisis peak, for $Lag > 20$, aligning with the DFA maximum and the GB crisis.

Table 1: Calculations underlying ROC curves

	Crisis Occurs	No Crisis Occurs
Crisis Signalled	(a) Correct	(b) Type II error
No Crisis Signalled	(c) Type I error	(d) Correct

See Figure 7 for an example of a ROC curve.

or high (e.g. 2.5%) level.²³

4.1.1. Receiver Operating Characteristics

The ROC approach, described by Fawcett (2004) and Jordà and Taylor (2011) has been widely applied in studies of credit cycles and systemic financial crises, including IMF (2011), Drehmann and Juselius (2014) and Detken et al. (2014). The underlying assumption is that the regulator operating the CyCB faces a binary choice for its level (0% or 2.5%.) Given a real-valued indicator y_t and the decision rule

$$\text{CyCB} = \text{“High”} \Leftrightarrow y_t > C$$

where $y_t > C$ is interpreted as a Crisis Signal. The regulator may search over possible values of C to produce a rule that guides their decisions of when to raise and lower the CyCB.

Together with our binary crisis indicators, this type of simple rule gives us only 4 possible outcomes as summarized in Table 1.

Outcomes (a) and (d) are preferred over the other two, which represent type I (outcome c) and type II (outcome b) errors. However, assigning relative values to these four cases (as would be required if we sought to choose C to minimise an expected loss function, for example) is problematic.²⁴ Instead, ROC analysis shows the usefulness of y_t by providing the set of potential trade-offs between Type I and Type II errors, expressed as a Hit rate

²³This abstracts from questions of how high or low the CyCB should be set, and so is relevant even for regulators choosing to set a positive “neutral” rate for the buffer. We return to this case below.

²⁴For example, it would require us to estimate the expected reduction in the overall economic costs of a systemic banking crisis conditional on having a higher value of the CyCB as well as gauge the expected economic costs of additional restrictions on bank capital in the absence of a systemic crisis.

(defined here as the fraction $a/(a+c)$) and a False Alarm rate (defined here as the fraction $b/(b+d)$).²⁵ We can trace out all feasible combinations of Hit rates and False Alarm rates by varying C over the full range of y_t . The resulting curve, shown in Figure 7, is called an ROC curve. High values of C imply that the CyCB will rarely be High, giving both a low Hit rate and a low False Alarm rate (in the lower left of the figure.) Conversely, low values of C imply that the CyCB will commonly be High, resulting in both higher Hit rates and False Alarm rates (in the upper right of the figure.)

If y_t contains no useful information about the onset of banking crises, then the CyCB would in effect be set randomly and we would expect ROC curve to lie along the main diagonal from $(0,0)$ to $(1,1)$. Indicators with ROC curves that lie above and the left of those of alternative are to be strictly preferred as they permit superior combinations of hit and False Alarm rates. In general, however, ROC curves of alternative indicators may cross, which prevents us from ranking their desirability without additional information. Despite this, a popular characterisation of ROC curves is the Area Under the Curve (AUC), which simply measures the surface area of the unit square which lies below the ROC. Curves with higher AUCs are associated with more desirable outcomes, while an AUC of 0.5 implies an indicator with no useful information.²⁶ Previous studies comparing different predictors of banking crises may principally focus on comparing their AUCs (such as Drehmann and Yetman, 2018; Drehmann and Yetman, 2021).

Early indications from the BCBS and subsequent statements from some national regulators have suggested that CCyBs should only be used “infrequently”, and that they are expected to be left at 0% or “Low” for the great majority of the time. In ROC analysis, if C is chosen so that a signal will only be given fraction p of the time and the ROC curve

²⁵The Hit rate is also referred to as the True Positive (TP) rate or the Sensitivity, while $1 -$ the False Alarm rate is also referred to as the selectivity or the True Negative (TN) rate.

²⁶Mason and Graham (2002) show that tests of the null hypothesis of equal AUCs are equivalent to Mann-Whitney U -statistics, but Robin et al. (2011) note that similar tests are not available for pAUCs. See also (Yang et al., 2024) for robust hypothesis tests and the relationship of AUC to other popular measures of concordance.

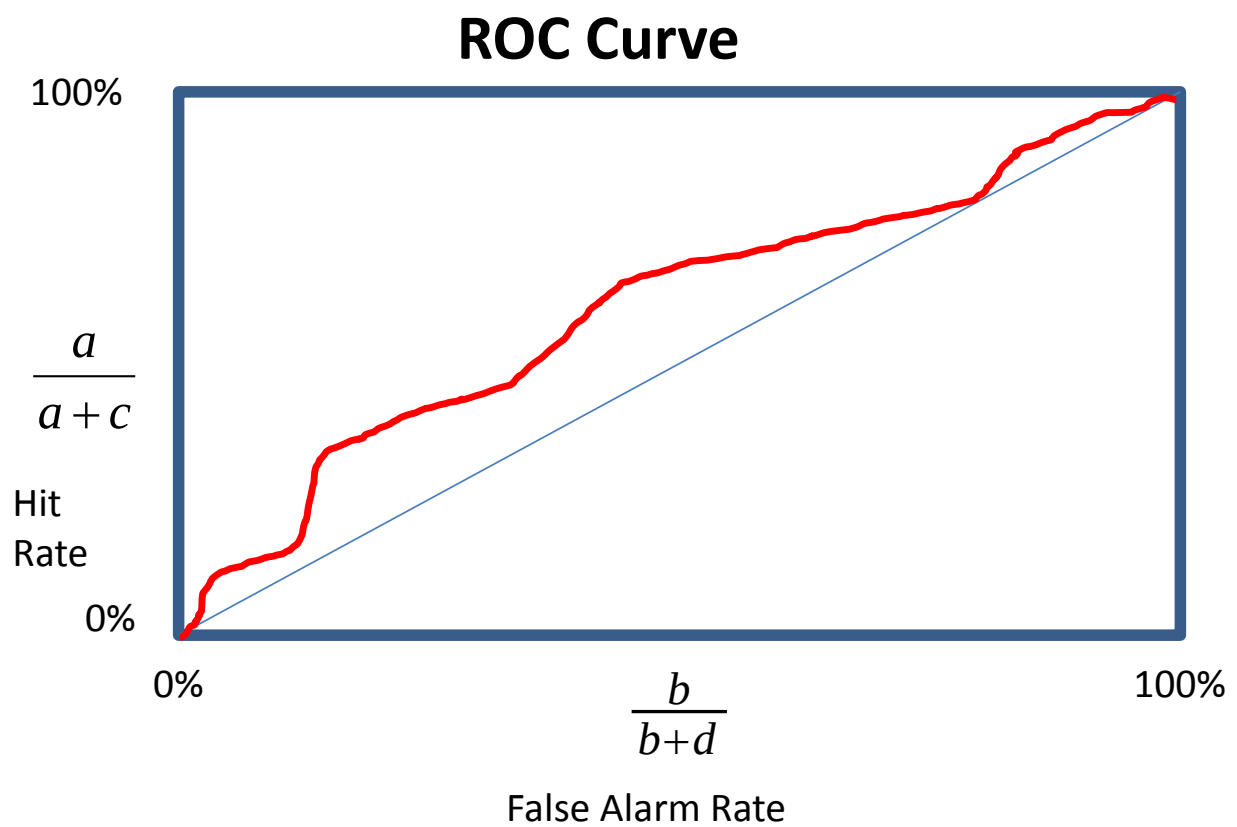


Figure 7: An ROC Curve
 See Table 1 for definitions of (a)-(d).

lies above the 45-degree line, then its False Alarm rate must be $< p$. This suggests that the portion of the curve that may be of greatest interest is the portion in the lower left corner. In the results we present below, we pay special attention to performance in the range $0 \leq p \leq 0.1$, calculating the area under the ROC curve in this region as the partial AUC (pAUCs).²⁷ This value is rescaled so that 1.0 is its maximum value and 0.5 corresponds to a statistic with no useful information, allowing an easy comparison to the overall AUC.

4.1.2. Prediction Timing and Scoring

The ROC analysis described above relies on the classification of actions into “successful” and “unsuccessful” outcomes. Our goal here is to gauge the degree to which a given measure y_t may help regulators in deciding when to *increase* the CyCB. We leave the important question of judging when to lower the CyCB in a systemic banking crisis to other research. Accordingly, in our analysis below, we omit the 8 quarters following the start of a banking crisis from all our calculations.²⁸

In addition, regulars seek to announce an increase in the CyCB many quarters in advance of a potential crisis.²⁹ We therefore score the predictions implied by y_t based on its ability to classify the crisis indicator I_{t+h} , which implies a forecast horizon of h periods.³⁰

4.2. Empirical Results

Figure 8 and Table 2 show results for the four Credit gap estimators discussed above using a forecast horizon $h = 5$ quarters. The table shows that overall the four different methods produced somewhat similar results in terms of AUC, ranging from a low of just

²⁷Results for the interval $0 \leq p \leq 0.25$ are available from the authors on request.

²⁸This is the same approach used by Drehmann and Juselius (2014), among others. Results are relatively robust to the precise number of quarters omitted. This reflects the facts that (1) crises are relatively rare, so this has little effect on the overall False Alarm rate, and (2) the minimum time span between crisis starts is considerably more than 8 quarters.

²⁹Among other factors, this reflects the practical need to provide financial institutions time to adjust to the new capital requirements.

³⁰We report results for selected values of h . Results reported by (Drehmann and Yetman, 2021), among others, show that AUCs are not overly sensitive to small changes in h .

Table 2: ROC Results for Four Gap Estimates

Gap	DFA_MSE	HP_1sided	Diff_20Q	Hamilton
AUC	0.749	0.717	0.708	0.739
pAUC ^a	0.604	0.629	0.577	0.593
p-Value ^b AUC	NA	0.003	0.001	0.424
p-Value pAUC	NA	0.005	0.007	0.257

^a pAUC is based on a False Alarm Rate over the [0.0, 0.1] interval.

^b p-Values reflect tests of the null hypothesis that the AUC (pAUC) for DFA_MSE is equal to that of another Gap estimate. These bootstrap p-values are approximate and should be interpreted with great caution.

over 70% for the 5Yr-Difference to a high of almost 75% for the DFA-MSE. Formal tests of the null hypothesis that the AUC of DFA-MSE are equal to the AUC of another gap estimate (see Table 2, “p-value AUC”) are rejected for all but the Hamilton gap, but we caution that these significance may be overstated.³¹ The error bars shown in Figure 8 provide similar 95% confidence intervals for the true positive rate at given values of the false positive rate for the DFA-MSE. At no point do they exclude more than one of the other curves, and no other curve ever lies above them.

The shape of the ROC curves also show differences that may be important for policymakers. In particular, the different gaps appear to have advantages over different false positive (“Alarm”) rates. For example, for rates of 10-40%, the DFA provides the highest true positive (“Hit”) rate, whereas for rates of 40-60% the Hamilton gap has an advantage. Perhaps surprisingly, while the HP gap performs poorly overall (and especially badly at high false positive rates), it is the best performing gap for false positive rates under 10%. In other

³¹The p-values are based on an stratified bootstrap. As such, it may fail to adequately consider the impact of both temporal dependence as well as cross-sectional clustering. (Yang et al., 2024) propose more robust methods that we hope to incorporate in a subsequent draft.

words, although the DFA gap has the best overall performance, if regulators were willing to impose “high” capital requirements roughly half the time, they might also wish to consider the Hamilton gap.

However, if regulators were instead quite reluctant to impose higher capital requirements, their choice of estimator could be quite different. Table 2 also shows the pAUCs for false positive rates of 10% or less (shown by the region of the ROC curves to the left of the grey vertical line in Figure 8). These show that the HP filter generally gives the best performance of the four estimators in this area, and may be statistically superior to all except the Hamilton gap. For example, these results imply that for regulators could still achieve a true positive rate of $> 30\%$ for a false positive rate of $< 4\%$ using the HP filter. Boosting this from 30 to 40% would require a false positive rate of near 10%, but could be done using either the HP or the DFA-MSE filter. These results suggest that regulators’ preferences may have a role to play in determining which gap estimators they should prefer.³²

We also think that the shape of the ROC curve offers another reason why some regulators in recent years have moved to adopt a “positive neutral value” of the CyCB, despite the original guidance from the BIS suggesting that the buffer should only be used “infrequently.” As Figure 8 and the above discussion makes clear, deploying the buffer infrequently implies accepting relatively low true positive rates.³³ Achieving a true positive rate of 50% implies accepting a false positive rate of about 14%, and a 70% rate raises this still further to at least 31%. None of the four methods that we examine are selective enough to enable regulators to be even mildly confident of deploying the CyCB in advance of a crisis unless it is frequently deployed.

³²Differences in their preferences for the frequency with which the CyCB is deployed may be related to such things as different evidence-based perceptions of the marginal cost of bank capital rather than the subjective preferences of the regulator.

³³As was explained above, this is because the frequency with which the buffer is deployed puts an upper bound on the false positive rate, and so constrains the true positive rate as the ROC curve illustrates.

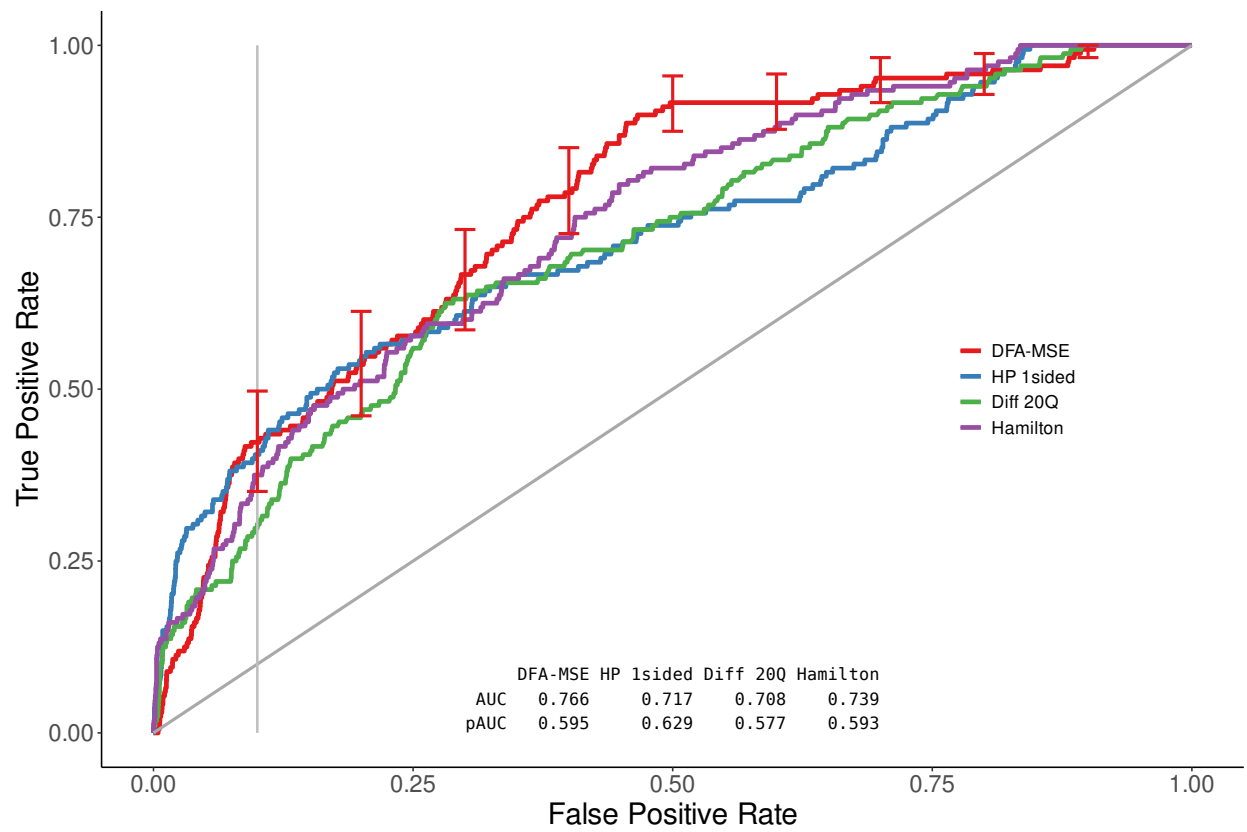


Figure 8: ROC Curves for Four Gap Estimates
The red error bars indicate an approximate 95% confidence interval around the indicated point on the ROC curve. The gray vertical line indicates the 10% cutoff for the pAUC.

4.2.1. Sensitivity Analysis

To better understand the robustness of our results, we also performed some sensitivity analysis. Appendix B.1 shows ROC curves using a forecast horizon for crises of 8 rather than 5 quarters. Consistent with the studies cited previously, this has no significant impact on our results. However, we found that our ROC analysis was sensitive to a largely neglected feature of credit gaps; that of scaling and centering. The remainder of this section explains why this is the case (and why it may have been neglected previously) and shows the impact of simple changes in scales or levels.

The idea that the scale (or the level) of a credit gap might affect the ROC curve (or the AUC) may seem counter-intuitive. This is because an indicator’s ROC curve (and therefore its AUC) depends only on the *ranking* of its values, not the cardinal values themselves. This implies that the ROC curve should be unaffected not only by the scale or the mean of the gap, but also by any monotonic transformation (such as e^{Gap} .)

For (bandpass) HP gaps, the mean is close to zero by construction; for lowpass filters, which capture growth in x_t (owing to $A(0) > 0$), the mean is generally non-zero. Crucially, however, both the scale and (for lowpass filters) the mean of the gaps vary substantially across countries. It is precisely this cross-country variation that distorts the rankings: when scales differ, a disproportionately small number of countries — those with the most volatile gaps — dominate the highest and lowest ranks (assuming similar means). When means differ, countries with larger mean gaps are overrepresented at the top of the ranking, while those with smaller means cluster at the bottom.

We demonstrate these effects in Figures 9 and 10. In the former, we rescale the gaps as $\tilde{y}_{j,t} = y_{j,t}/\sqrt{\text{Var}(y_{j,t})}$: this transformation homogenizes the scale across countries while making more visible cross-country differences in mean levels, particularly for lowpass designs. Mean-level effects weigh most heavily on the DFA, whose passband is the narrowest among the filters considered, making the mean level a more dominant driver of the gap³⁴. Once scale

³⁴Non-zero means were most important for the DFA-MSE and the 5Yr Difference gaps. 13 of the 43 nations had mean DFA-MSE gaps > 1 , 3 had means > 2 ; for 5Yr Difference gaps, 8 had means > 1 .

and level effects are stripped out, countries contribute more uniformly to the ranking, with representation becoming appreciably more balanced across both tails of the distribution. In their presence, however, scale or level effects can randomly steepen or flatten the ROC curve at the outset, depending on which outlier country initiates the curve — either enabling or undermining accurate crisis detection: in the former case, the ROC rises steeply, yielding a larger pAUC; in the latter, it tracks close to (or under) the diagonal, yielding a smaller pAUC.

On these grounds, we recommend harmonizing scale and level effects across countries before computing ROC curves. This reduces noise by preventing one or a few "outlier" countries — those with disproportionately large scales or (absolute) mean levels — from exerting undue influence on ROC estimates, especially in the lower tail (which governs early warning performance) and, to a somewhat lesser extent, in the upper tail. Standardization provides a natural and intuitive way to achieve this noise reduction, and we therefore argue that Figure 10 — corresponding to the "Standardized" rows in Table 3 — offers the most representative characterization of mean filter performance across countries³⁵. Under this specification, differences between filters emerge as statistically significant for certain pairings and across specific regions of the ROC, cf. Figure 10.

5. Conclusions

How should credit cycles be measured and how should their predictive content for financial crises be evaluated? Measuring credit cycles relies on filtering techniques to extract relevant features from the data. We analyse two filter characteristics, amplitude and time-shift, when filters are applied in levels versus first differences. The first-difference analysis yields sharper distinctions at cycle- and lowest-frequencies: all filters except the HP filter act as lowpass filters with non-zero amplitude at frequency zero, whereas the HP filter suppresses the zero frequency entirely yet exhibits the smallest time-shift. These properties give rise to a trade-off between spurious cyclicalities and lead time at cycle peaks: while the HP

³⁵While standardization is appropriate for lowpass filters, bandpass filters may only require scaling.

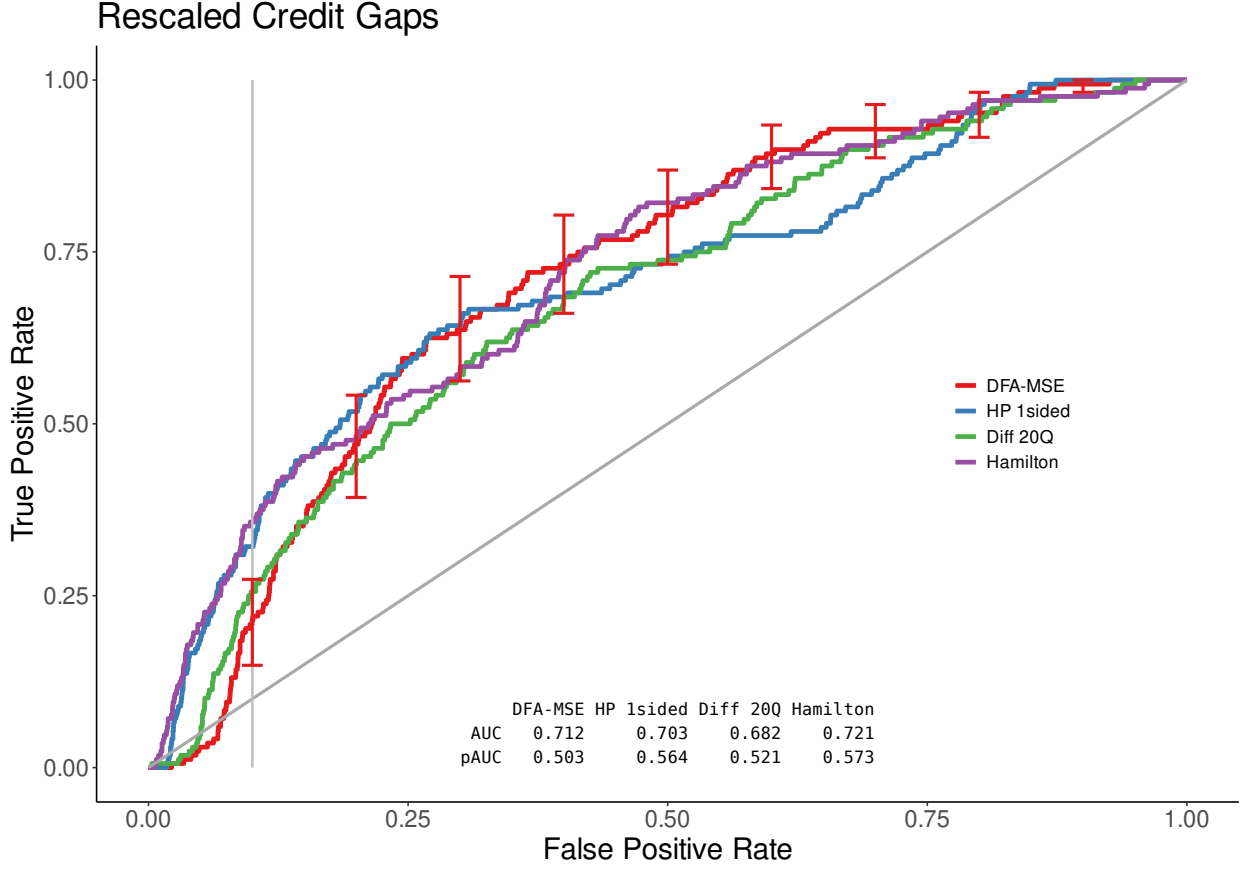


Figure 9: ROC Curves for Rescaled Gap Estimates

The gap measure $y_{j,t}$ for country j was transformed so that the rescaled measure $\tilde{y}_{j,t} = y_{j,t}/\sqrt{\text{Var}(y_{j,t})}$.

The red error bars indicate an approximate 95% confidence interval around the indicated point on the ROC curve. The gray vertical line indicates the 10% cutoff for the pAUC.

filter tends to lead at genuine crisis cycles, it may max out at spurious cycles either before or after the actual crisis. Depending on prevailing priorities, either aspect of this trade-off may warrant greater emphasis. One way to navigate this tension is to tailor the low-frequency passband characteristics—specifically, the amplitude at frequency zero and the width of the passband. Alternatively, one can combine filters that occupy different positions along this trade-off.

We find DFA to achieve the highest overall AUROC. But along the ROC curve, rankings depend on policymakers' tolerance for false alarms and can vary when rescaled or standardized gaps are considered. More fundamentally, irrespective of filter choice, none of the four

Table 3: Sensitivity of AUC Results

Gap	DFA_MSE	HP_1sided	Diff_20Q	Hamilton
AUC:				
Base Case ^a	0.749	0.717	0.708	0.739
Rescaled ^b	0.712	0.703	0.682	0.721
Standardized ^c	0.771	0.724	0.699	0.722
pAUC: ^d				
Base Case ^a	0.604	0.629	0.577	0.593
Rescaling ^b	0.503	0.564	0.521	0.573
Standardized ^c	0.610	0.599	0.575	0.573

^a These are the same values as those reported in Table 2

^b The gap measure $y_{j,t}$ for country j was transformed so that the rescaled measure $\tilde{y}_{j,t} = y_{j,t} / \sqrt{\text{Var}(y_{j,t})}$.

^c Each gap series for each nation was standardized to $\mu = 0$, $\sigma = 1$.

^d pAUC is based on a False Alarm Rate over the $[0.0, 0.1]$ interval.

methods we investigate can deliver simultaneously timely and selective warnings—hit rates remain below 50 percent for low false alarm rates—pointing to a binding constraint on what credit gap measurement can realistically achieve.

This constraint has implications for policy. No filter-based signal can be made reliably selective under an infrequent-deployment norm, and a framework conditioning buffer build-up on such a signal will either frequently miss crises or impose capital requirements that often prove unnecessary. The positive neutral CCyB—which maintains a standing buffer releasable in periods of stress rather than depending on a predictive activation signal—sidesteps this dilemma. Our findings provide a rationale for the shift toward positive neutral CCyBs, pursued by numerous countries in recent years.

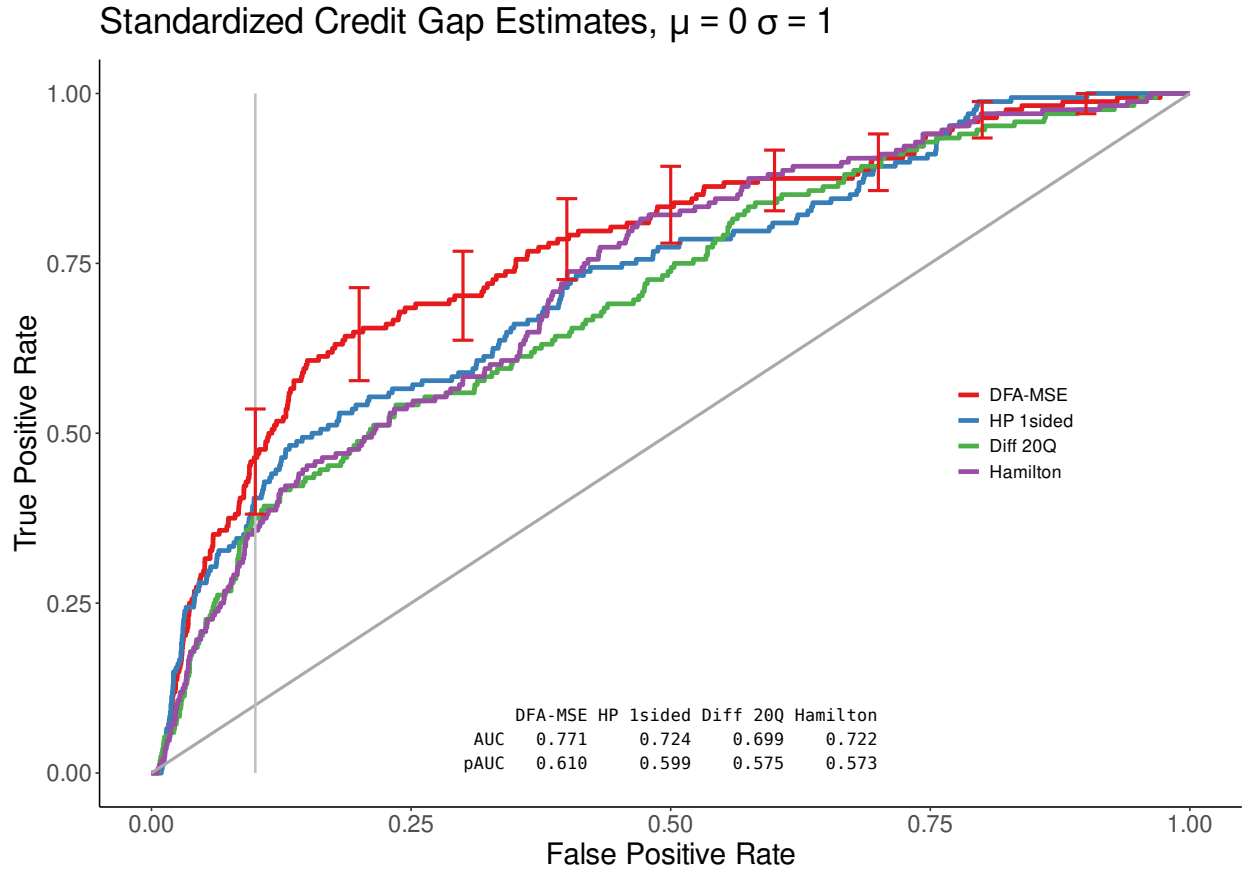


Figure 10: ROC Curves for Standardized Gap Estimates

The three nations with the highest values for each gap, as well as the three nations with the lowest, were dropped from the analysis. The red error bars indicate an approximate 95% confidence interval around the indicated point on the ROC curve. The gray vertical line indicates the 10% cutoff for the pAUC.

References

- Afanasyeva, E. (2020), ‘Can forecast errors predict financial crises? Exploring the properties of a new multivariate credit gap’, *FEDS Working Paper 2020-045*.
- Alessandri, P., Bologna, P., Fiori, R. and Sette, E. (2015), ‘A note on the implementation of the countercyclical capital buffer in Italy’, *Bank of Italy Occasional Paper 278*.
- Baba, C., Dell’Erba, S., Detragiache, E., Harrison, O., Mineshima, A., Musayev, A. and Shahmoradi, A. (2020), ‘How should credit gaps be measured? An application to European countries’, *IMF Working Paper WP/20/6*.
- BCBS (2010), ‘Guidance for national authorities operating the countercyclical capital buffer’, *Basel Committee on Banking Supervision, BIS*.

- Behn, M., Detken, C., Peltonen, T. A. and Schudel, W. (2013), ‘Setting countercyclical capital buffers based on early warning models: Would it work?’, *ECB Working Paper No. 1604* .
- Beltran, D. O., Jahan-Parvar, M. R. and Paine, F. (2021), ‘Optimizing credit gaps for predicting financial crises: Modelling choices and tradeoffs’, *FRB International Finance Discussion Paper* (1307).
- Bernanke, B. S., Gertler, M. and Gilchrist, S. (1999), ‘The financial accelerator in a quantitative business cycle framework’, *Handbook of macroeconomics* **1**, 1341–1393.
- Beutel, J., List, S. and von Schweinitz, G. (2019), ‘Does machine learning help us predict banking crises?’, *Journal of Financial Stability* **45**, 100693.
- Borio, C. E. V. and Lowe, P. (2002), ‘Asset prices, financial and monetary stability: Exploring the nexus’, *BIS Working Paper No. 114* .
- Brave, S. A. and Lopez, J. A. (2019), ‘Calibrating macroprudential policy to forecasts of financial stability’, *International Journal of Central Banking* **15**(1).
- Chen, M. S. and Svirydzenka, K. (2021), ‘Financial cycles—early warning indicators of banking crises?’, *IMF Working Paper No. 2021/116* .
- Detken, C., Weeken, O., Alessi, L., Bonfim, D., Boucinha, M., Castro, C., Frontczak, S., Giordana, G., Giese, J., Wildmann, N., Kakes, J., Klaus, B., Lang, J. H., Puzanova, N. and Welz, P. (2014), ‘Operationalising the countercyclical capital buffer: Indicator selection, threshold identification and calibration options’, *ESRB: Occasional Paper Series* (2014/5).
- Drehmann, M., Borio, C. E. V. and Tsatsaronis, K. (2011), ‘Anchoring countercyclical capital buffers: the role of credit aggregates’, *International Journal of Central Banking* **7**(4), 189–240.
- Drehmann, M. and Juselius, M. (2014), ‘Evaluating early warning indicators of banking crises: Satisfying policy requirements’, *International Journal of Forecasting* **30**(3), 759–780.
- Drehmann, M. and Yetman, J. (2018), ‘Why you should use the Hodrick-Prescott filter - at least to generate Credit Gaps’, *BIS Working Paper No. 744* .
- Drehmann, M. and Yetman, J. (2021), ‘Which credit gap is better at predicting financial crises? A comparison of univariate filters’, *International Journal of Central Banking* **17**(70), 225–255.
- ECB (2017), ‘Financial stability review’, **May 2017**.
- ECB/ESRB (2025), ‘Using the countercyclical capital buffer to build resilience early in the cycle’, *Joint ECB/ESRB report on the use of the positive neutral CCyB in the EEA* .
- Edge, R. M. and Meisenzahl, R. (2011), ‘The unreliability of credit-to-GDP ratio gaps in real-time: Implications for countercyclical capital buffers’, *International Journal of Central Banking* **7**(4), 261–298.
- ESRB (2014), ‘Recommendation of the European Systemic Risk Board of 18 June 2014 on guidance for setting countercyclical buffer rates’, *Official Journal of the European Union, European Systemic Risk Board* (2014/C 293/01).

- Fawcett, T. (2004), ‘ROC Graphs: Notes and practical considerations for researchers’, *Machine Learning* **31**, 1–38.
- Galán, J. (2019), ‘Measuring credit-to-GDP gaps. the Hodrick-Prescott filter revisited’, *The Hodrick-Prescott Filter Revisited (May 8, 2019)*. Banco de Espana Occasional Paper (1906).
- Hamilton, J. D. (2018), ‘Why you should never use the Hodrick-Prescott filter’, *Review of Economics and Statistics* **100**(5), 831–843.
- Hamilton, J. D. and Leff, D. (2020), ‘Measuring the credit gap’, *mimeo* .
- Hartl, T. (2025), ‘Robust trend estimation for strongly persistent data with unobserved memory’, *Journal of Business & Economic Statistics* (forthcoming), 1–23.
- Hartwig, B., Meinerding, C. and Schüller, Y. S. (2021), ‘Identifying indicators of systemic risk’, *Journal of International Economics* **132**, 103512.
- IMF (2011), Global financial stability report, Technical report, International Monetary Fund. Chapter 3.
- Jokipii, T., Nyffeler, R. and Riederer, S. (2021), ‘Exploring BIS credit-to-GDP gap critiques: The Swiss case’, *Swiss Journal of Economics and Statistics* **157**(7).
- Jordà, O. and Taylor, A. M. (2011), Performance evaluation of zero net-investment strategies, Working Paper 17150, National Bureau of Economic Research.
- Kindleberger, C. (1978), *Manias, panics and crashes: A history of financial crises*, New York: Basic Books. Revised edition, 1989; 3rd edition, 1996, Wiley and Sons; 4th revised edition with Robert Aliber, 2000, Wiley and Sons.
- Laeven, L. and Valencia, F. (2018), ‘Systemic banking crises revisited’, *IMF Working Paper WP/18/206* .
- Laeven, L. and Valencia, F. (2020), ‘Systemic banking crises database II’, *IMF Economic Review* **68**, 307–361.
- Lang, J. H. and Welz, P. (2017), ‘Measuring credit gaps for macroprudential policy’, *Financial Stability Review* **1**, 144.
- Mason, S. J. and Graham, N. E. (2002), ‘Areas beneath the relative operating characteristics (ROC) and relative operating levels (ROL) curves: Statistical significance and interpretation’, *Quarterly Journal of the Royal Meteorological Society* **128**(584), 2145–2166.
- McElroy, T. (2008), ‘Exact formulas for the Hodrick-Prescott filter’, *The Econometrics Journal* **11**(1), 209–217.
- Minsky, H. P. (1977), A theory of systematic fragility, in Altman and Sametz, pp. 138–152.
- Nguyen, T. C., Castro, V. and Wood, J. (2022), ‘A new comprehensive database of financial crises: Identification, frequency, and duration’, *Economic Modelling* **108**, 105770.
- Phillips, P. C. B. and Jin, S. (2015), ‘Business cycles, trend elimination, and the HP filter’, *Cowles Foundation Discussion Paper No. 2005* .

- Quast, J. (2026), ‘Beyond endpoints: Revisiting the challenges of estimating trends in real time’, *mimeo*, <http://dx.doi.org/10.2139/ssrn.4177293> .
- Reinhart, C. M. and Rogoff, K. S. (2009), ‘The aftermath of financial crises’, *American Economic Review* **99**(2), 466–472.
- Repullo, R. and Salas, J. S. (2011), ‘The countercyclical capital buffer of Basel III: A critical assessment’, *CEPR Discussion Paper No. DP8304* .
- Robin, X., Turck, N., Hainard, A., Tiberti, N., Lisacek, F., Sanchez, J.-C. and Müller, M. (2011), ‘pROC: an open-source package for R and S+ to analyze and compare ROC curves’, *BMC Bioinformatics* **12**, 77.
- Schularick, M. and Taylor, A. M. (2012), ‘Credit booms gone bust: Monetary policy, leverage cycles, and financial crises, 1870–2008’, *American Economic Review* **102**(2), 1029–1061.
- Schüler, Y. S. (2018), ‘Detrending and financial cycle facts across G7 countries: Mind a spurious medium term!’, *ECB Working Paper Series No. 2138* .
- Schüler, Y. S. (2020), ‘On the credit-to-GDP gap and spurious medium-term cycles’, *Economics Letters* **192**(109245).
- van Norden, S. and Wildi, M. (2015), ‘Basel III and the prediction of financial crises’, *Prepared for Swiss National Bank Conference, mimeo* .
- Wezel, T. (2019), ‘Conceptual issues in calibrating the Basel III countercyclical capital buffer’, *IMF Working Paper No. 2019/086* .
- Wildi, M. (2026a), ‘The accuracy smoothness dilemma in prediction: a novel multivariate forecast approach’, *Journal of Time Series Analysis (accepted for publication)* .
- Wildi, M. (2026b), ‘Sign accuracy, mean-squared error and the rate of zero crossings: a generalized forecast approach’, *arXiv:2601.06547* .
- Wildi, M. and McElroy, T. (2016), ‘Optimal real-time filters for linear prediction problems’, *Journal of Time Series Econometrics* **8**(2), 155–192.
- Wildi, M. and McElroy, T. S. (2019), ‘The trilemma between accuracy, timeliness and smoothness in real-time signal extraction’, *International Journal of Forecasting* **35**(3), 1072–1084.
- Yang, L., Lahiri, K. and Pagan, A. (2024), ‘Getting the ROC into Sync’, *Journal of Business & Economic Statistics* **42**(1), 109–121.

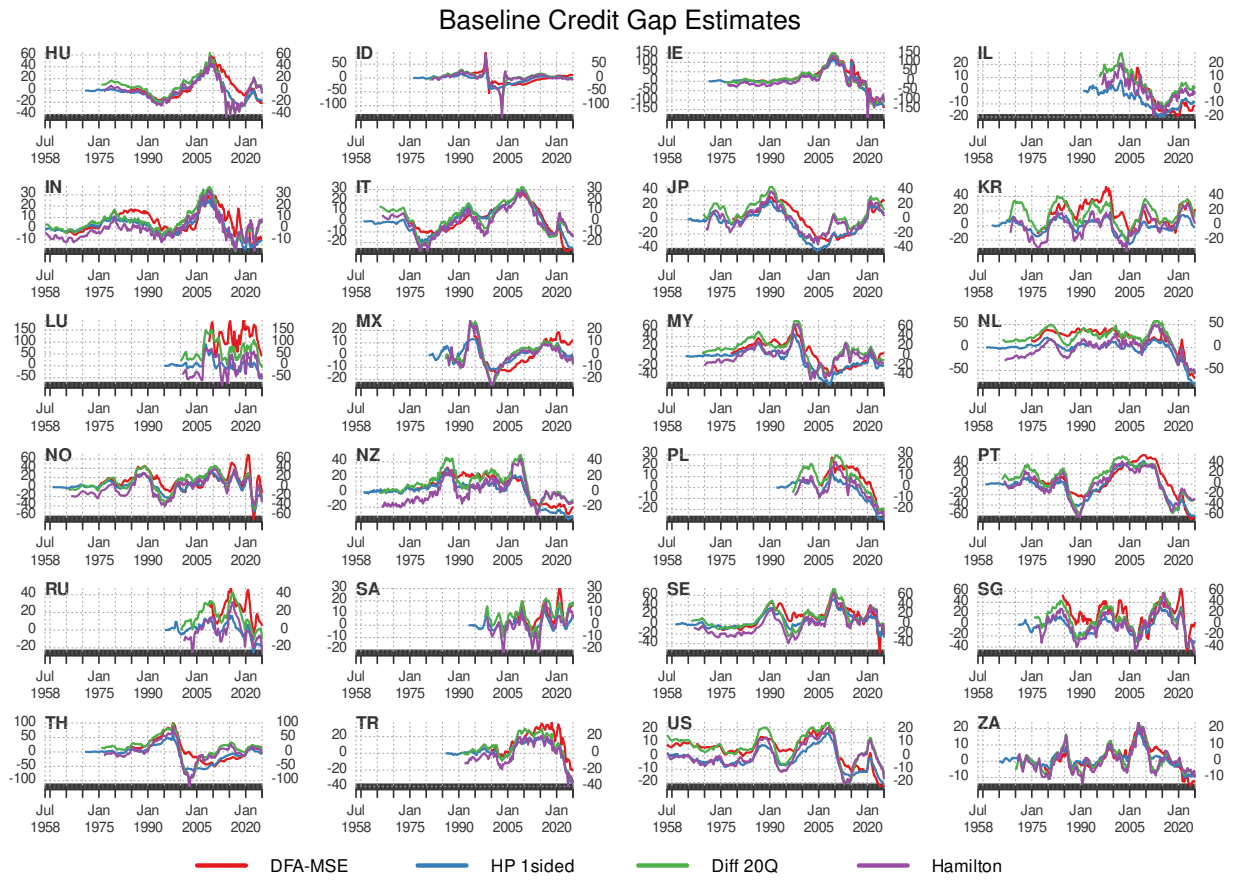


Figure A.12: Gap Estimates: DFA-MSE, HP(400K), 5Yr Diff., Hamilton (cont.)

the other three, it often tends to be smoother and shows the most independent variation. This broadly reflects the distinctive amplitude function that it possesses, as we saw above.

Third, in a few cases, we see that it estimates a credit gap that is persistently above zero (see BR, CL, CO, HK, LU and RU.) This is consistent with the property (noted previously in Section 3) of filters that are low-pass in differences; with sufficiently strong growth in the credit ratio, they need not return to zero and thus may avoid spurious cyclicalities.

Appendix B. Sensitivity Analysis

Appendix B.1. Robustness to Forecast Horizon

The ROC analysis presented above used a 5Q forecast horizon, and we noted previous research found that results were relatively insensitive to modest changes in forecast horizon. Below, Figure Appendix B.1 presents results for an 8Q forecast horizon. It shows results that are quantitatively very similar to those above.

Appendix B.2. Trimming vs Scaling

We also examined an alternative to the rescaling and standardization of the estimated credit gaps that we examined in the body of the paper. Instead, we tried to reduce the influence of extreme values by considering a reduced set of countries, eliminating those nations with the 3 highest and 3 lowest gap estimates for each of our four estimators.³⁶ The results (see Figure B.14) show that the AUCs of all four gap estimators and most of and pAUCs fall somewhat and to varying degrees. However, the decline is not as pronounced as it was when rescaled gaps were used instead. We interpret this as evidence that the scaling effect is not confined to a small number of countries, but that it is not as robust a method as standardization.

³⁶Because the set of countries selected varied slightly from estimator to estimator, this caused us to drop 8 of our 43 countries: ES, HK, ID, IE, LU, NL, PT, and TH. For every estimator, IE (Ireland) had the lowest and the highest or 2nd highest gap values.

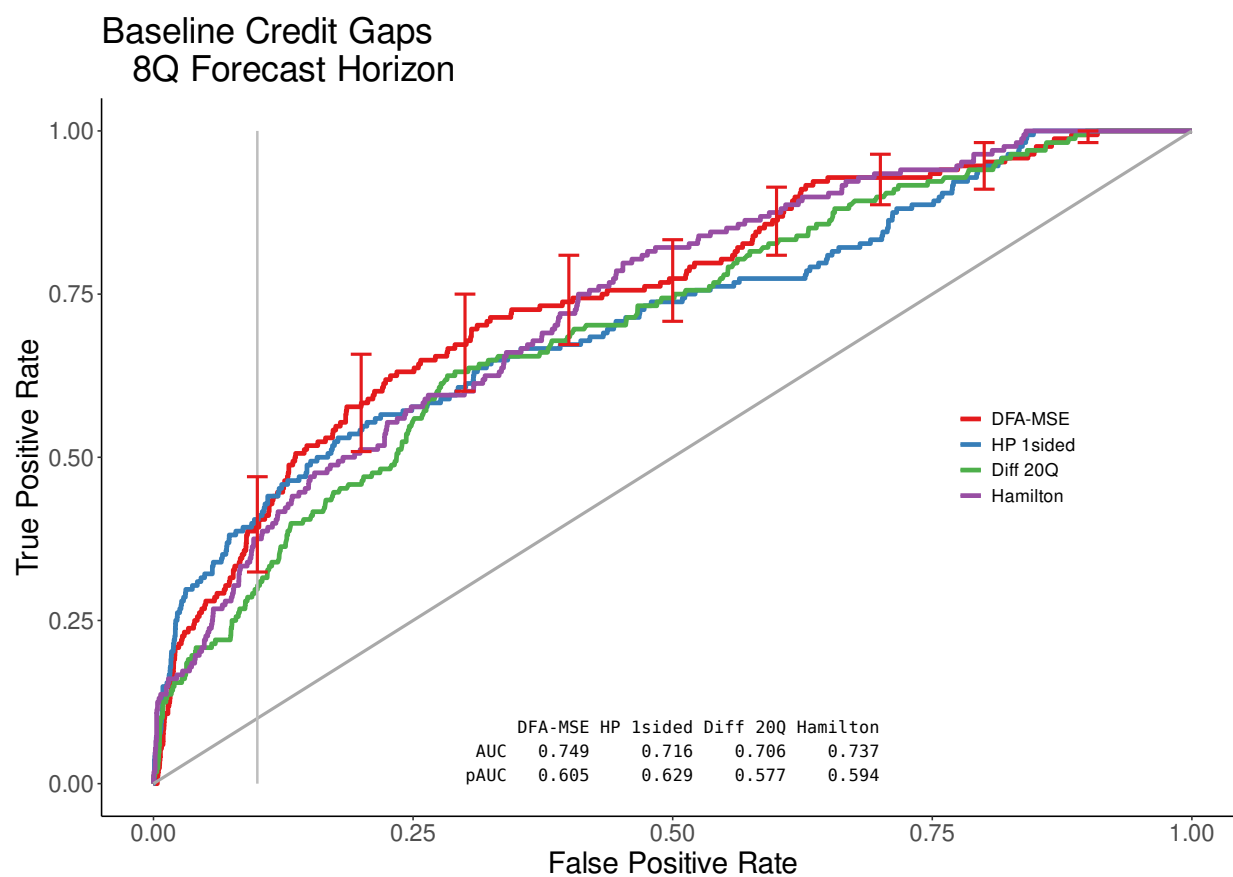


Figure B.13: ROC Curves: 8Q Forecast Horizon

The red error bars indicate an approximate 95% confidence interval around the indicated point on the ROC curve. The gray vertical line indicates the 10% cutoff for the pAUC.

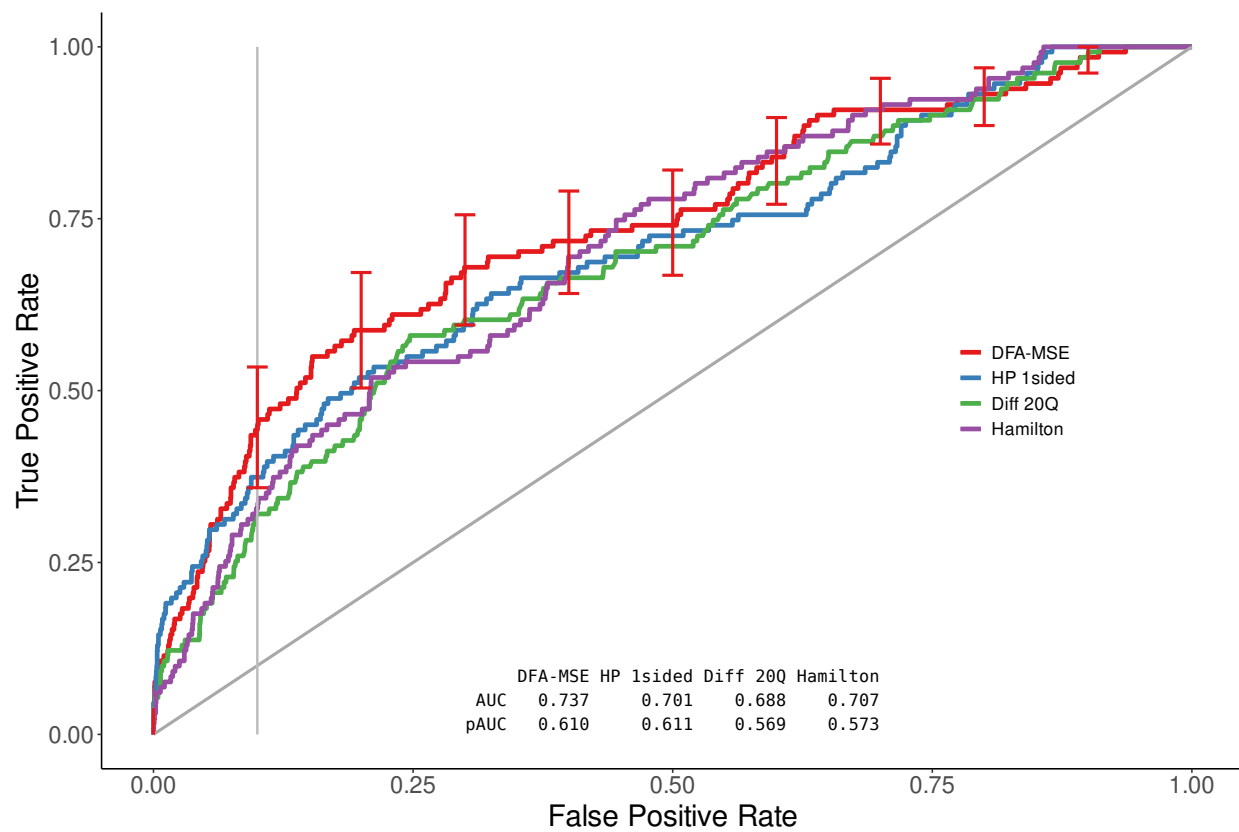


Figure B.14: ROC Curves for Trimmed Country Sample

The three nations with the highest values for each gap, as well as the three nations with the lowest, were dropped from the analysis. The red error bars indicate an approximate 95% confidence interval around the indicated point on the ROC curve. The gray vertical line indicates the 10% cutoff for the pAUC.